Retail Business Performance & Profitability Analysis

Project Type: Exploratory data analysis **Dataset:** Practice retail dataset (150 orders)

1. Introduction

Retail businesses use sales and profit data to understand product performance, customer demand, and regional trends. This project explores a sample retail dataset to summarize sales performance and identify patterns that can support business decisions.

2. Abstract / Objective

The objective was to inspect and analyze retail order data using Python. The analysis covered dataset structure, missing values, duplicate rows, total sales and profit, sales by category and region, customer segments, discount levels, product subcategories, and monthly sales trends. The dataset is synthetic and intended for learning; results should not be treated as real business findings.

3. Tools Used

Google Colab, Python, Pandas, and Matplotlib. Pandas was used to load, inspect, group, aggregate, and export data. Matplotlib was used to create bar and line charts.

4. Steps Involved

- Loaded the CSV file into a Pandas DataFrame.
- Inspected rows, columns, data types, missing values, and duplicate rows.
- Converted Order_Date to a datetime data type.
- Calculated total sales, total profit, units sold, average order sales, and overall profit margin.
- Grouped sales and profit by category, subcategory, region, customer segment, discount, and month.
- Created category-level bar charts and a monthly sales trend line chart.
- Exported the analyzed dataset to a CSV file.

5. Key Findings

In the practice dataset, total profit was 17,479.61, total quantity sold was 634 units, average sales per order was 520.88, and overall profit margin was approximately 22.37%. Category, region, customer segment, discount, product, and month comparisons were explored in Colab. Add the specific top-performing categories and months from your saved notebook outputs.

6. Conclusion

This project demonstrates a basic retail analytics workflow: data inspection, date conversion, aggregation, visualization, and CSV export. It builds foundational skills in Pandas and Matplotlib. The analysis could be extended with SQL queries and an interactive dashboard. Because the dataset is synthetic and small, conclusions are suitable for practice rather than operational decisions.